VCU College of Engineering

Project #CS309 and AI Recommender Tool

First Report

Prepared for

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Capital One

By

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Executive Summary

Organizations everywhere are increasingly focused on Governance, Risk, and Compliance to protect data, secure integrity, and meet regulatory requirements. In large operational companies, the process of risk management for new technology projects is often slowed by manual, time consuming research. This can lead to significant delays in project initiation and inconsistent risk assessments.

The main problem comes from the outdated, manual risk assessment process that requires GRC specialists to evaluate the entire project and roles through long spreadsheets and handbooks. With this slow method, there is potential for inaccurate risk categorization which can expose Capital One to avoidable risks. For a faster, clear, and more user-friendly solution, we can design an automated tool that simplifies this process while ensuring consistency and compliance.

The goal of the AI recommender tool is to design and prototype an AI powered chatbot that simplifies the risk assessment of new projects and roles. The chatbot will accept user input description and title and interact with the user to provide a risk classification of low, medium, or high based on general industry GRC policies. Ultimately, this tool will make the risk assessment more efficient, faster, and accessible.

Key Objectives:

- Developing a chatbot with user friendly and appealing interface for a quick assessment
- Ensuring that the tool can generate and provide risk classifications quickly
- Using generalized GRC principles and industry knowledge without relying on formulas

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Section A. Problem Statement

In this fast-paced digital workplace across all industries, we have a greater focus on Governance, Risk, and Compliance (GRC) to safeguard data, secure operational integrity, and meet regulatory requirements. In large technology-based organizations, it is difficult to effectively maintain risk management due to the amount and complexity of the many projects deployed. The problem occurs constantly whenever a new technology project is proposed. For a large technology-driven organization, this could mean the manual assessment process is initiated hundreds or even thousands of times per year across different departments. The issue isn't a one-time event but a recurring bottleneck.

The main problem comes from the current manual and time-intensive process used to assess the risk of a new technology project. Previously, this process involved project owners completing lengthy questionnaires, consulting complex internal risk handbooks, and navigating complex risk calculation spreadsheets to determine a project's risk categorization. This leads to time consumption and delay, where the process can have significant delays in project initiation, as risk assessments may require extensive effort and multiple rounds of review by GRC specialists. And for non-specialists, there is a high barrier to getting started, as they may find the existing tools and documents difficult to use, which can discourage them from being proactive about risks. Whether analyzed by a GRC specialist or non-specialist, purely relying on manual interpretation of complex rules and mandates can lead to inconsistent risk ratings or subjectivity across similar projects or project owners. The manual system does not scale effectively with the exponential growth in new technology projects. This can create a potential backlog and increase organizational exposure to unmitigated risk. The unmet engineering need is for a fast and user-friendly automated tool that can instantly provide an initial project risk level by referencing core GRC principles, which normalizes the initial risk assessment process and frees up GRC specialists for more complex tasks.

The problem primarily impacts technology project managers, product owners, development teams, and direct users, as well as the GRC and compliance teams who must review these assessments. The project client is Capital One, and the key stakeholder is Ashish Kulkami.

There are many potential costs associated with a deficient risk assessment process, like economic costs with fines and penalties for regulatory non-compliance, and costs associated with system breaches or failures. There are also societal/reputational costs with loss of customer trust following a data incident resulting from an under-assessed risk. This project falls within the general field of Regulatory Technology, specifically applying Natural Language Processing or Generative AI techniques to automate GRC processes.

The primary goal of this project is to design and prototype a chatbot that accepts a brief technology project description, which then prompts a simple Q&A, and outputs a clear risk classification of low, medium, or high based on general industry GRC knowledge. This project contributes to the field by demonstrating a Proof-of-Concept for replacing form-based or spreadsheet-driven GRC intake with an intuitive, chat-based interface. By abstracting the complexity of internal GRC mandates and relying on generalized industry standards due to the

client's formulas, the project showcases how AI and conversational interfaces can significantly advance GRC efficiency, speed, and user experience.

Existing solutions to technology risk assessment range from proprietary enterprise GRC Platforms (like ServiceNow, Archer, MetricStream, orDiligent HighBond) to simpler custom tools. While commercial platforms offer risk registers and control libraries, they often require extensive configuration and are not currently designed for fast risk intake. This project seeks to build on the concept of automated risk scoring by prioritizing user experience and its quick response, unlike a data-entry model, to a conversational one.

Section B. Engineering Design Requirements

This section describes the goals and objectives of the project, as well as all **realistic constraints** to which the design is bound. It is meant to provide a structure that helps to formulate the problem. Design requirements are often derived from client or stakeholder needs. They may consider benchmarking against or improving on currently available solutions, providing novel techniques or design solutions, integration with existing components, systems, or equipment, required codes and standards, general observations of the problem space, etc. Describe how the requirements provided below were researched and decided upon. Common design requirements often include considerations of the design efficacy, cost, safety, reliability, usability, and risk, among others.

Note: The design requirements should be revisited between major reports to ensure that the design objectives and constraints still accurately reflect the client needs and project goals and to make sure that the team is on track to meet all goals and objectives.

Note: The codes and standards section is not required for the Project Proposal, but is required for all subsequent reports. This section should be comprehensive and thorough, requiring a significant research effort.

B.1 Project Goals (i.e. Client Needs)

Our goal is to create a simple, student-built chatbot that can automatically figure out the risk level of a technology project (Low, Medium, or High) without using complicated spreadsheets. The idea is to make something fast, smart, and easy to use that helps identify if a project meets common tech and data laws or if there's something risky about.

Project Goals:

- Build a chat-style bot that takes in a short title or description and gives back a clear risk level (Low, Medium, and High).
- Let the bot ask a few extra questions if it needs more info (Kind of like answering "Who, What, When, Where, and Why" but in a quick and simple way).
- Follow general industry rules (Since we cannot use any company's private formulas)
- Use basic risk management knowledge that applies to most tech projects, focusing on whether projects follow legal and safety requirements and not on the individuals themselves.

B.2 Design Objectives

Our design focuses on keeping things clear, creative, and functional. The objective should be SMART, Specific, measurable, achievable, relevant, and time bound.

- UI design: Make a chat-style interface where users can type in short responses, and the bot replies with questions and results.
- Performance: Make sure the bot gives a full risk summary and checklist in 5 seconds or less for normal cases.

B.3 Design Specifications and Constraints

A list of design specifications and constraints include all limitations, restrictions, and requirements of the design. They are firm limits that must be met for a design to be acceptable and are ultimately used to measure the success of a design. Each specification or constraint should map to one or more design objective(s) and explicitly state *how the design* will meet the objectives. Specifications and constraints should be specific and are often numerical. They must be measurable or testable to prove that the design has met all of the design objectives. Numerical metrics may include qualifying statements such as “at least,” “at most,” “between,” “exactly” or include a set of discrete values. Avoid subjective, untestable constraints (e.g. “environmentally friendly”, “user friendly”, “nice looking”, etc.).

Realistic constraints can come take on a variety of forms including accessibility, aesthetics, codes, constructability, cost, ergonomics, extensibility, functionality, interoperability, legal considerations, maintainability, manufacturability, marketability, policy, regulations, schedule, standards, sustainability, or usability. Examples of physical constraints might include numerical limits or ranges on overall size envelope, weight, pressures, stresses, flow rates, voltages, current, power consumption, hardware limitations, data constrains, interoperability, etc. Other constraints might include production unit cost, expected part/device life, or maintenance requirements.

Some general examples of constraints are as follows:

- Design must integrate with existing equipment or process (Functional constraint) – examples include specific piping or electrical connections, available size envelope, power requirements, etc.
- Design must not exceed some overall weight (Functional constraint) – specify weight limits or range
- Design must not exceed some power consumption or operate at some efficiency (Functional constraint) – examples include current, voltage, power requirements, mechanical or electrical efficiency
- Must be cheaper to produce than currently available designs (Cost constraint) – provide specific cost limit or range
- Design must be made from readily available materials (Manufacturability constraint) – provide exact materials
- Design must operate for so many hours without requiring maintenance (Maintainability constraint) – provide exact time or range

- Design must operate within some hardware with specifications up to certain values (Hardware constraint) – provide exact hardware specifications
- Design must operate within some constraints on data collection and processing (Data Constrains) – provide exact guidelines or restrictions
- Design must operate within a set of systems or platforms requiring compatibility with specific protocols or APIs (Interoperability Constrains) – provide specific systems of platforms.

Note: Relevant codes and standards should be incorporated into the design specifications and constraints listed above.

Section C. Scope of Work

The project scope defines the boundaries of the project encompassing the key objectives, timeline, milestones and deliverables. It clearly defines the responsibility of the team and the process by which the proposed work will be verified and approved. A clear scope helps to facilitate understanding of the project, reduce ambiguities and risk, and manage expectations. In addition to stating the responsibilities of the team, it should also explicitly state those tasks which fall *outside* of the team's responsibilities. *Explicit bounds* on the project timeline, available funds, and promised deliverables should be clearly stated. These boundaries help to avoid *scope creep*, or changes to the scope of the project without any control. This section also defines the project approach, the development methodology used in developing the solution, such as waterfall or agile (shall be chosen in concert with the faculty advisor and/or project sponsor). Good communication with the project sponsor and faculty advisor is the most effective way to stay within scope and make sure all objectives and deliverables are met on time and on budget.

C.1 Deliverables

The project delivers an AI-based recommender tool that will be used by employers for a quick, concise description of a role that abides by the law. The following deliverables are agreed upon with the project sponsor and faculty advisor:

- Technical Deliverables:
    - System design documentation: High-level architecture diagrams that outline data flow, user interaction, and logic.
    - Design analyses: Summary of system testing, accuracy metrics, and performance analysis.
    - Computer code: Frontend and backend code connected to the project that provides optimal performance and time complexity.
    - User manual: Recommendations for usage and future maintenance.
- Academic Deliverables:
    - Team Contract
    - Project Proposal
    - Preliminary design report
    - Fall poster
    - Presentation
    - Final Design Report
    - Capstone EXPO poster
    - Capstone EXPO presentation

In order to mitigate risks associated with the completion and delivery of the project deliverables, provide an outline of the most potentially disruptive, foreseeable obstacles. Some important issues to discuss with the design team, sponsor, and faculty advisor include the following:

- Deliverables that consist of tangible results like poster printing and prototype will require campus access in the workshop. At least two members of the team should be available with regular campus access to physically complete these tasks.
- Coding, documentation, and model training can be completed remotely with occasional online calls for communication. Collaboration tools like Google Drive, Github, and Zoom will be used during this process. Resources like Python, Jupyter Notebooks, and Google Colab will be used to ensure that remote work will be completed effectively.
- This project may require access to APIs, pretrained models, or external datasets. The team will identify, confirm, and secure these earlier into the semester to prevent licensing or potential delays. If, for any reason, hardware resources are needed and there is a delay, the team will use cloud-based computing alternatives to maintain workflow continuity.

C.2 Milestones

Phase	Task	Start Date	End Date
Phase 1: Planning	Project Setup & Team Formation	8/28/2025	9/1/2025
	Project Definition & Initial Research	9/1/2025	9/25/2025
	Client/Stakeholder Interviews	9/25/2025	10/5/2025
	Initial Requirements & Restraints Definition	10/5/2025	10/15/2025
	Competitive Analysis & Benchmarking	10/15/2025	11/10/2025
	Prototype High-Level Design & Architecture	11/10/2025	12/1/2025
	Draft: Project Proposal (Report)	12/1/2025	12/15/202 Milestone
	Final Project Proposal Revision	1/4/2026	1/11/2026
	Detailed Risk Logic Formula Design	1/11/2026	2/4/2026
	GRC Knowledge Base Implementation	2/4/2026	2/23/2026
	Chatbox UI/UX Development (Front-End)	2/10/2026	3/24/2026
	Core NLP/AI Integration	2/15/2026	3/24/2026

Phase 2: Design & Implementation	Risk Calculation Engine Development	3/10/2026	4/15/2026
	Draft: Mid-Semester Design Report	3/24/2026	4/5/2026 Milestone
Phase 3: Testing & Expo	Internal Unit & Integration Testing	4/5/2026	4/18/2026
	Client/Stakeholder Feedback Loop	4/8/2026	4/28/2026
	Final Prototype Verification	4/28/2026	5/3/2026 Milestone
	Draft: Senior Design Abstract	4/28/2026	5/3/2026
	Draft: Expo Poster & Presentation	4/20/2026	5/3/2026
	Final Design Report Compilation	4/28/2026	5/14/2026 Milestone
	Final Preparation & Rehearsal	5/6/2026	5/14/2026
	Senior Design Expo	5/6/2026	5/6/2026 Milestone

C.3 Resources

Most resources we need we should be able to get for free ourselves, such as Visual Studio Code. It would be helpful if test data could be provided (example data containing no real life info).

Section D. Concept Generation

Section E. Concept Evaluation and Selection

Section F. Design Methodology

F.1 Computational Methods (e.g. FEA or CFD Modeling, example sub-section)

F.2 Experimental Methods (example subsection)

F.3 Architecture/High-level Design (example subsection)

F.5 Validation Procedure

Section G. Results and Design Details

G.1 Modeling Results (example subsection)

G.2 Experimental Results (example subsection)

G.3 Prototyping and Testing Results (example subsection)

G.4. Final Design Details/Specifications (example subsection)

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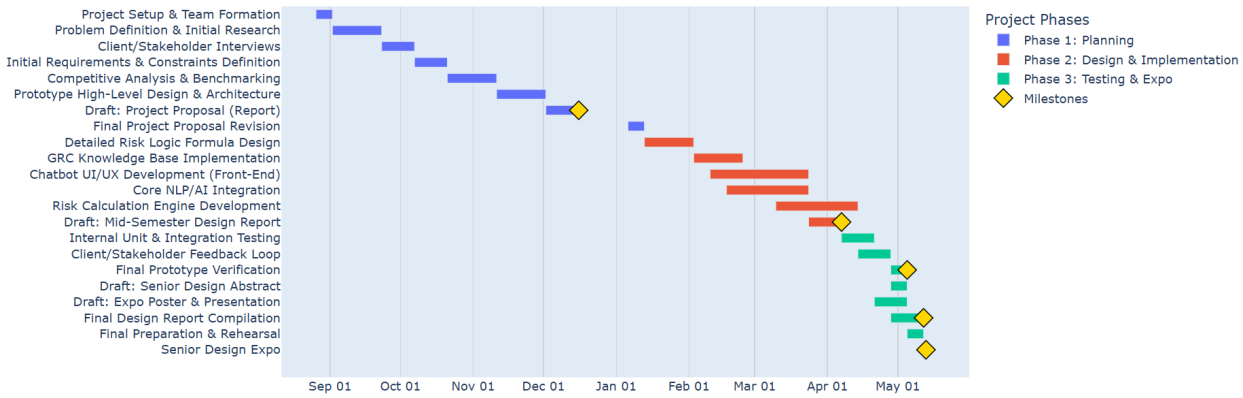
H.7. Ethical Considerations

Section I. Cost Analysis

Section J. Conclusions and Recommendations

Appendix 1: Project Timeline



Appendix 2: Team Contract (i.e. Team Organization)

Step 1: Get to Know One Another. Gather Basic Information.

Task: This initial time together is important to form a strong team dynamic and get to know each other more as people outside of class time. Consider ways to develop positive working relationships with others, while remaining open and personal. Learn each other's strengths and discuss good/bad team experiences. This is also a good opportunity to start to better understand each other's communication and working styles.

We weren't able to meet with Dr. Leonard yet, as he wants to wait until he is able to contact all of the sponsors before meeting with students. Once we meet with him, we will make any changes necessary (reminder: check with professor during class to see if we should resubmit this after changes have been made).

Team Member Name	*Strengths each member bring to the group*	*Other Info*	*Contact Info*
Jake Carduck	Hard working, open-minded to ideas, happy to learn new skills	I have worked on a project involving using OCR on PDF documents before.	carduckj@vcu.edu
Michelle Koh	Dedicated to learn, flexible, empathetic	I tend to be away from my phone once I am home	kohme@vcu.edu
Kelly Hwangpo	Avid learner, good communication, open-minded/unbias	I may overcommunicate and push to get things done earlier due to complicated schedule	hwangpoks@vcu.edu
Dylan Tran	Good communication skills, hard working, and empathetic	Hard to reach sometimes, understanding	trandn3@vcu.edu

Other Stakeholders	*Notes*	*Contact Info*
John Leonard		*jdleonard@vcu.edu*
Lakshmi Narayana Rasalay		*lakshminarayana.rasalay@capitalone.com*

Step 2: Team Culture. Clarify the Group's Purpose and Culture Goals.

Task: Discuss how each team member wants to be treated to encourage them to make valuable contributions to the group and how each team member would like to feel recognized for their efforts. Discuss how the team will foster an environment where each team member feels they are accountable for their actions and the way they contribute to the project. These are your Culture Goals (left column). How do the students demonstrate these culture goals? These are your Actions (middle column). Finally, how do students deviate from the team's culture goals? What are ways that other team members can notice when that culture goal is no longer being honored in team dynamics? These are your Warning Signs (right column).

Resources: More information and an example Team Culture can be found in the Biodesign Student Guide "Intentional Teamwork" page (webpage | PDF)

Culture Goals	***Actions***	***Warning Signs***
Being on time to every meeting	*- Set up meetings on Discord* *- Send reminder message in Discord on day of meeting* *- Put the meeting in your calendar*	*- Student misses first meeting, without excuse* *- Student misses meetings afterwards – issue is brought up with faculty advisor*
Informing the group of any delays in completing assignments	*- Stay up to date with each other's project responsibilities* *- Set reasonable deadlines and note when an extension is needed*	*- Student shows up for weekly meeting with no considerable work done* *- Fails to update the group until the deadline has passed*
Enhanced Team Collaboration	*- Communicate, ask questions* *- Set specific goals for the week/make a deadline* *- Listen to everyone, even if they are not necessarily right*	*- Student misses a deadline or submits poor/wrong work* *- Student stays quiet during meeting and not giving any feedback*

Step 3: Time Commitments, Meeting Structure, and Communication

Task: Discuss the anticipated time commitments for the group project. Consider the following questions (don't answer these questions in the box below):

- What are reasonable time commitments for everyone to invest in this project?
- What other activities and commitments do group members have in their lives?
- How will we communicate with each other?
- When will we meet as a team? Where will we meet? How Often?
- Who will run the meetings? Will there be an assigned team leader or scribe? Does that position rotate or will same person take on that role for the duration of the project?

Required: How often you will meet with your faculty advisor advisor, where you will meet, and how the meetings will be conducted. Who arranges these meetings? See examples below.

Meeting Participants	***Frequency*** ***Dates and Times / Locations***	***Meeting Goals*** ***Responsible Party***
Students Only	*Every Thursday @ 6:00PM, On Discord Voice Channel*	*Stand up meeting: Update group on day-to-day challenges and accomplishments* *(Michelle will record these for the weekly progress reports and meetings with advisor)*
Students Only	*Every Tuesday @ 4:45pm at Engineering Research Building*	*Actively work on project* *(Michelle will document these meetings by taking photos of whiteboards, physical prototypes, etc, then post on Discord and update Capstone Report)*
Students + Faculty advisor	*As needed through Zoom. Ideally on Tuesdays @ 4:45 PM or Advisor's office (conference room if they can reserve)*	*Update faculty advisor and get answers to our questions* *(Michelle will scribe; Dylan will create meeting agenda and lead meeting)*
Project Sponsor	*First and Third Friday of the month @ 3PM* *If sponsor is available, we'll figure out Zoom or in person details* *If not, then we'll update the sponsor via email.*	*Update project sponsor and make sure we are on the right track (Michelle will scribe; Dylan will create meeting agenda and lead meeting; Dylan will present prototype so far)*

Step 4: Determine Individual Roles and Responsibilities

Task: As part of the Capstone Team experience, each member will take on a leadership role, ***in addition to*** contributing to the overall weekly action items for the project. Some common leadership roles for Capstone projects are listed below. Other roles may be assigned with approval of your faculty advisor as deemed fit for the project. For the entirety of the project, you should communicate progress to your advisor specifically with regard to your role.

- **Before meeting with your team**, take some time to ask yourself: what is my "natural" role in this group (strengths)? How can I use this experience to help me grow and develop more?
- **As a group,** discuss the various tasks needed for the project and role preferences. Then assign roles in the table on the next page. Try to create a team dynamic that is fair and equitable, while promoting the strengths of each member.

Communication Leaders

Suggested: Assign a team member to be the primary contact for the client/sponsor. This person will schedule meetings, send updates, and ensure deliverables are met.

Suggested: Assign a team member to be the primary contact for faculty advisor. This person will schedule meetings, send updates, and ensure deliverables are met.

Common Leadership Roles for Capstone

1. **Project Manager:** Manages all tasks; develops overall schedule for project; writes agendas and runs meetings; reviews and monitors individual action items; creates an environment where team members are respected, take risks and feel safe expressing their ideas.
2. **Logistics Manager:** coordinates all internal and external interactions; lead in establishing contact within and outside of organization, following up on communication of commitments, obtaining information for the team; documents meeting minutes; manages facility and resource usage.
3. **Financial Manager:** researches/benchmarks technical purchases and acquisitions; conducts pricing analysis and budget justifications on proposed purchases; carries out team purchase requests; monitors team budget.
4. **Systems Engineer:** analyzes Client initial design specification and leads establishment of product specifications; monitors, coordinates and manages integration of sub-systems in the prototype; develops and recommends system architecture and manages product interfaces.
5. **Test Engineer:** oversees experimental design, test plan, procedures and data analysis; acquires data acquisition equipment and any necessary software; establishes test protocols and schedules; oversees statistical analysis of results; leads presentation of experimental finding and resulting recommendations.
6. **Manufacturing Engineer:** coordinates all fabrication required to meet final prototype requirements; oversees that all engineering drawings meet the requirements of machine shop or vendor; reviews designs to ensure design for manufacturing; determines realistic timing for fabrication and quality; develops schedule for all manufacturing.

Team Member	***Role(s)***	***Responsibilities***
Jake Carduck	*Technical Lead*	- *Breaking up the project into distributable segments* - *Quality control (including comments in code).*
Michelle Koh	Documentation Lead	- Keep up with weekly documentation - Scribe meeting notes
Kelly Hwangpo	External Communications/Logistics Manager	- Communicate with faculty member and sponsor - Discuss any potential finances with sponsor
Dylan Tran	Project Organizer/Lead Presenter	- *Ensure the team stays aligned with project goals.* - *Prepare and coordinate presentation materials.*

Step 5: Agree to the above team contract

Team Member: Jake Carduck *Signature:* Jake Carduck

Team Member: Michelle Koh *Signature:* Michelle Koh

Team Member: Kelly Hwangpo *Signature:* Kelly Hwangpo

Team Member: Dylan Tran *Signature:* Dylan Tran

References

Provide a numbered list of all references in order of appearance using APA citation format. The reference page should begin on a new page as shown here.

[1] VCU Writing Center. (2021, September 8). *APA Citation: A guide to formatting in APA style.* Retrieved September 2, 2024. https://writing.vcu.edu/student-resources/apa-citations/

[2] Teach Engineering. *Engineering Design Process*. TeachEngineering.org. Retreived September 2, 2024. https://www.teachengineering.org/populartopics/designprocess